

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder (BALSAM-2)
Short Title/Acronym: BALSAM-2 **Protocol Number:** CN012-0037 **NCT Number:** NCT06951711 **Sponsor Name:** Bristol-Myers Squibb
Investigational Product: KarXT (xanomeline-trospium) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol-Myers Squibb
Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of KarXT compared to placebo in treating symptoms of mania during a 3-week inpatient period. **Secondary Objectives:** To assess the severity, improvement, and efficacy of treatment using the Clinical Global Impressions-Bipolar (CGI-BP) scale, and to evaluate the safety and tolerability of KarXT.

2.2 Background & Rationale KarXT is an investigational muscarinic receptor agonist that has demonstrated long-term efficacy and safety benefits in treating schizophrenia. This study aims to evaluate whether KarXT can be effectively utilized to treat and manage manic episodes in individuals with Bipolar-I Disorder, potentially offering a novel treatment option that avoids the typical side effect profiles of standard, existing psychiatric therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **\$N\$:** 274 **Number of Sites:** Multicenter **Estimated Study Start:** 13-Jun-2025 **Estimated Primary Completion:** 02-Nov-2026 **Estimated Study Completion:** 02-Nov-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Primary diagnosis of Bipolar-I disorder established by a comprehensive psychiatric evaluation. 2. Experiencing an acute episode or relapse of mania or mania with

mixed features (≤ 3 weeks) that requires hospitalization. 3. Young Mania Rating Scale (YMRS) score of ≥ 20 and Clinical Global Impressions-Bipolar (CGI-BP) ≥ 4 at Screening and at Baseline. 4. Ages 18 to 65 years, inclusive, and all genders.

4.2 Exclusion Criteria 1. Any primary DSM-5-TR disorder other than BP-I with mania or mania with mixed features within 12 months before screening (e.g., rapid cycling with ≥ 4 distinct mood episodes in one year). 2. At risk for suicidal behavior at screening or baseline as determined by the Investigator's clinical assessment and the Columbia-Suicide Severity Rating Scale (C-SSRS). 3. Cirrhosis, liver cancer, clinically significant liver disease, or use of psychotropic medications that were not successfully washed out in no more than 14 days prior to the first dose. 4. DSM-5-TR diagnosis of moderate to severe substance use disorder (except tobacco use disorder) within the 12 months before screening, or current use as determined by urine toxicology screen or alcohol test.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT (flexible dosing) vs. Placebo, administered daily. **Route of Administration:** Oral **Duration of Treatment:** 3-week double-blind inpatient treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard non-psychotropic medications for comorbidities as prescribed by the investigator. **Prohibited:** All psychotropic medications must be washed out in no more than 14 days prior to the first dose of the study drug.

6. Study Timeline & Visit Schedule

The study schedule is divided into four main phases. During the **Screening** visit (Day -14 to -1), activities include the washout of psychotropics, comprehensive psychiatric evaluation, YMRS, CGI-BP, and C-SSRS scoring, alongside obtaining Informed Consent. At **Baseline** (Day 0), participants undergo Randomization and Inpatient admission, with YMRS & CGI-BP confirmation prior to the First Dose. The **Interim** phase (Weeks 1-2) consists of continuous Inpatient monitoring, efficacy assessments, and safety labs. The **End of Study** phase (Week 3 to 7) begins with Final inpatient efficacy assessments (Week 3), followed by an off-treatment safety follow-up period up to week 7.

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to -1	Washout of psychotropics, psychiatric evaluation, YMRS, CGI-BP, C-SSRS, and Informed Consent.
Baseline	Day 0	Randomization, Inpatient admission, YMRS & CGI-BP confirmation, First Dose.
Interim		

Weeks 1-2 | Inpatient monitoring, efficacy assessments, and safety labs. | | **End of Study** | Week 3 to 7 | Final inpatient efficacy assessments (Week 3), followed by an off-treatment safety follow-up period up to week 7. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy of KarXT compared to placebo in treating symptoms of mania. **Measurement Method:** Measured by the change in the Young Mania Rating Scale (YMRS) total score (comprising 11 items assessing severity of manic symptoms) during the 3-week inpatient period.

7.2 Secondary & Exploratory Endpoints - Change in severity and improvement using the Clinical Global Impressions-Bipolar (CGI-BP) scale. - Safety and tolerability measures (incidence of adverse events, safety labs, and C-SSRS scores).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient monitoring by clinical staff for expected procholinergic and anticholinergic AEs, routine safety laboratory tests (safety labs), as well as regular C-SSRS assessments for suicidal ideation. **Serious Adverse Events (SAEs):** Reported promptly within site awareness according to standard sponsor protocols. **Data Monitoring Committee (DMC):** External safety monitoring board designated by Bristol-Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for adverse events. **Sample Size Justification:** Powered to detect a statistically significant difference in YMRS score reduction between KarXT and placebo during the trial timeline. **Handling of Missing Data:** Standard handling for dropouts (e.g., Mixed-Effect Model Repeated Measure [MMRM] or Multiple Imputation). **Oversight:** The Lead Statistician will oversee the execution of the Statistical Analysis Plan.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring visits throughout the active inpatient phase. **Audit Trail:** Data tracking and case report forms via validated electronic data capture (EDC) systems.

11. Study Identifiers & Governance

Primary ID: CN012-0037 Registry Number: NCT06951711 Other Identifiers: 2024-520195-94 (EU CTR), U1111-1316-7466 (WHO) Sponsor/Lead Organization: Bristol-Myers Squibb Study Title: BALSAM-2 Official Regulatory Title: A Phase 3, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Efficacy and Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder (BALSAM-2)

12. Clinical Framework (Bipolar I Disorder Specific)

Primary Condition: Bipolar-I Disorder, Manic Episodes Diagnosis Threshold: YMRS score ≥ 20 and CGI-BP ≥ 4 Medical Methodology: Comprehensive psychiatric evaluation (DSM-5-TR) Optimization Target: Reduction in manic symptoms and mood stabilization

13. Study Procedures & Methodology

Management Pathway: Inpatient psychiatric care and continuous monitoring Education Protocol (HCP): Clinical rater training and calibration for subjective scales (YMRS, CGI-BP, C-SSRS) Education Protocol (Patient): Inpatient trial procedures, informed consent, and self-reporting Quality Audit Mechanism: Centralized rater review and rigorous source data verification

14. Observation & Follow-Up Schedule

Total Duration: Up to 7 weeks (including screening, 3-week treatment, and safety follow-up) Onsite Visit Cadence: Continuous observation during the 3-week inpatient stay Remote Monitoring Frequency: N/A (Inpatient setting) Checklist Review Interval: Routine, standardized symptom assessments throughout the treatment phase

15. Sign-off & Approval

This report requires the final signature and approval date from the designated Principal Investigator, the Bristol-Myers Squibb Clinical Operations Lead, and the Lead Statistician.

Role	Name	Signature	Date				
Principal Investigator	[Name]	_____	06-Apr-2026				
Clinical Operations Lead	[Name]	_____	06-Apr-2026				
Lead Statistician	[Name]	_____	06-Apr-2026				